

1. Administrative & Study Identification

Full Study Title: A Phase III, Double-Blind, Multicenter, Randomized Study of Atezolizumab (Anti-PD-L1 Antibody) Versus Placebo as Adjuvant Therapy in Patients With High-Risk Muscle-Invasive Bladder Cancer Who Are ctDNA Positive Following Cystectomy **Short Title/Acronym:** IMvigor011 **Protocol Number:** B042843 **NCT Number:** NCT04660344 **Posting ID:** 150505505335117374 **Trial Status:** Active, Not Recruiting (ACTIVE_NOT_RECRUITING) **Sponsor Name:** F. Hoffmann-La Roche Ltd / Genentech / Roche **Investigational Product:** Atezolizumab (Tecentriq) / Placebo **Phase of Development:** Phase III **Medical Monitor:** F. Hoffmann-La Roche Ltd Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of atezolizumab compared with placebo in improving investigator-assessed disease-free survival (DFS) as adjuvant therapy in patients with high-risk MIBC who are ctDNA positive post-cystectomy. **Secondary Objectives:** To evaluate overall survival (OS), Independent Review Facility (IRF)-assessed DFS, disease-specific survival, safety, tolerability, and quality of life.

2.2 Background & Rationale To address the high relapse rates in patients with muscle-invasive bladder cancer (MIBC) post-surgery. Despite curative intent with radical cystectomy and neoadjuvant chemotherapy, approximately 50% of patients relapse within 2 years. Monitoring circulating tumor DNA (ctDNA) post-surgery to detect molecular residual disease (MRD) allows for the precise identification of patients at the highest risk of recurrence. This trial validates a personalized, ctDNA-guided approach, providing adjuvant atezolizumab only to patients who test positive for ctDNA, improving clinical outcomes while sparing MRD-negative patients from unnecessary treatment toxicity.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 2:1 (Atezolizumab : Placebo)

3.2 Planned Enrollment & Duration Target \$N\$: 761 patients enrolled in the surveillance phase **Number of Sites:** Global, Multicenter **Estimated Study Start:** 06-Apr-2021 **Current Status:** Active but not currently recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status of ≤ 2 and a life expectancy of ≥ 12 weeks. 2. Histologically confirmed muscle-invasive urothelial carcinoma (MIUC/MIBC) of the bladder, having undergone radical cystectomy. A minimum of 6 weeks must have elapsed from surgery, with full recovery and enrollment within 24 weeks. 3. TNM classification (based on AJCC Cancer Staging Manual, 8th Edition) of ypT2-4a or ypN+ and M0 (if prior neoadjuvant chemotherapy [NAC] was received) OR pT2-4a or pN+ and M0 (if prior NAC was not received and patient is cisplatin-ineligible, refused, or will not receive it based on physician's decision). 4. Blood plasma evaluation showing ctDNA positive status (defined as ≥ 2 mutations out of 16 identified via WES report). 5. Tumor PD-L1 expression per IHC evaluable by central testing. 6. Absence of residual disease and metastasis confirmed by a negative baseline CT or MRI scan of the pelvis, abdomen, and chest ≤ 28 days prior to randomization. 7. For women of childbearing potential: agreement to remain abstinent or use contraception and refrain from donating eggs.

4.2 Exclusion Criteria 1. Received prior adjuvant chemotherapy or post-surgical radiation therapy for UC, or any approved anti-cancer therapy within 3 weeks prior to randomization. 2. Malignancies other than UC within 5 years prior to study enrollment. 3. Active or history of autoimmune disease (Exception: autoimmune-related hypothyroidism on a stable dose of hormone replacement or controlled Type I diabetes on a stable insulin regimen are permitted). 4. Active infection with Hepatitis B (HBV), Hepatitis C (HCV), or HIV. *Exceptions:* Patients with positive HIV test are eligible if stable on ART, CD4 count $\geq 200/\mu\text{L}$, and undetectable viral load. Past/resolved HBV (HBV DNA negative) and HCV antibody-positive (HCV RNA PCR negative) patients are eligible. 5. Active tuberculosis confirmed by a test performed within 3 months prior to treatment initiation. 6. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or active pneumonitis on screening chest CT. 7. Significant cardiovascular disease (e.g., NYHA Class II or greater, myocardial infarction within 3 months, unstable arrhythmias or angina).

5. Intervention & Treatment Plan

5.1 Investigational Regimen **Investigational Drug:** Atezolizumab (Trade Name: Tecentriq) **ATC Code:** L01FF05 **EMA Product Information:** [EMA URL](#) **Dosage/Frequency:** Atezolizumab 1680 mg or matching Placebo, administered every 4 weeks. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 12 cycles (1 year), or until disease recurrence, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care, hormone-replacement therapy, oral contraceptives, and concomitant medications for co-morbidities as per investigator discretion. **Prohibited:** Concurrent administration of other systemic anti-cancer therapies, investigational agents (within 28 days or 5 half-lives prior to randomization), live/live-attenuated vaccines, and systemic immunosuppressants.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening (Surveillance)** | Weeks 6-24 post-op |
Informed consent, baseline CT/MRI imaging, and regular blood draws for ctDNA monitoring. | | **Baseline** | Day 0 | Randomization upon first ctDNA+ test, First IV Dose, Baseline Safety Labs. | | **Interim** | Every 4 weeks | IV Infusion of Atezolizumab/Placebo, Safety Labs, Efficacy Assessment. | | **End of Study** | Week 48 (Cycle 12) | Final Dose, Exit Interview, Safety Follow-Up, continued imaging for DFS/OS tracking. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Investigator-assessed Disease-Free Survival (DFS): Time from randomization to the first occurrence of a DFS event, defined as any of the following: * Local (pelvic) recurrence of urothelial carcinoma (including soft tissue and regional lymph nodes) * Urinary tract recurrence of urothelial carcinoma (including all pathological stages and grades) * Distant metastasis of urothelial carcinoma * Death from any cause

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS), defined as time from randomization to death from any cause. 2. Independent Review Facility (IRF)-Assessed DFS. 3. Investigator-Assessed Disease-Specific Survival. 4. Quality of Life assessment using the Bladder Cancer Muscle-Invasive 30 Questionnaire (EORTC QLQ-BLM30), a 30-question tool developed by the European Organisation for Research and Treatment of Cancer to assess QoL in patients with muscle-invasive bladder cancer. 5. Safety, tolerability, and ctDNA clearance rate.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection at every visit, graded according to NCI CTCAE parameters. **Serious Adverse Events (SAEs):** Reported to the sponsor within standard 24-hour timelines of investigator awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (iDMC) oversees the safety and efficacy data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary efficacy analyses; Safety Set for all patients receiving at least one dose of the study drug. **Sample Size Justification:** The study was powered to detect a statistically significant and clinically meaningful improvement in DFS and OS for the atezolizumab arm over the placebo arm among the randomized ctDNA-positive patients (2:1 ratio). **Handling of Missing Data:** Standard survival analysis methods including Kaplan-Meier survival estimates and Cox proportional hazards regression models, with appropriate right-censoring for time-to-event outcomes.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence, safety, and data integrity. **Audit Trail:** Validated Electronic Data Capture (EDC) system with secure audit trails and eCRF locking mechanisms.

11. Study Identifiers & Governance

Primary ID: B042843 **Registry Number:** NCT04660344 **Posting ID:** 150505505335117374 **Sponsor/Lead Organization:** F. Hoffmann-La Roche Ltd / Roche **Study Title:** IMvigor011 **Official Regulatory Title:** A Phase III, Double-Blind, Multicenter, Randomized Study of Atezolizumab (Anti-PD-L1 Antibody) Versus Placebo as Adjuvant Therapy in Patients With High-Risk Muscle-Invasive Bladder Cancer Who Are ctDNA Positive Following Cystectomy

12. Clinical Framework (Disease Specific)

Primary Condition: High-Risk Muscle-Invasive Bladder Cancer (MIBC) **Diagnosis Threshold:** Molecular residual disease (MRD) identified by a positive ctDNA test (≥ 2 mutations via WES assay) **Medical Methodology:** ctDNA-guided Adjuvant Immunotherapy **Optimization Target:** Disease-Free Survival (DFS) and Overall Survival (OS) extension

13. Study Procedures & Methodology

Management Pathway: Targeted Adjuvant Immune Checkpoint Inhibitor Pathway for MIBC **Education Protocol (HCP):** Site initiation visits and continuous training on precise ctDNA sample logistics and assay interpretation **Education Protocol (Patient):** Informed consent process covering whole exome sequencing, regular blood draws, QLQ-BLM30 quality of life assessments, and immunotherapy side effects **Quality Audit Mechanism:** Centralized Independent Review Facility (IRF) for radiographic imaging, DFS assessment, and specialized laboratory validation for ctDNA

14. Observation & Follow-Up Schedule

Total Duration: Up to 21 months of surveillance, 12 months of active treatment, and long-term survival follow-up **Onsite Visit Cadence:** Every 4 weeks during the 12-cycle treatment phase **Remote Monitoring Frequency:** Regular interval check-ins per the sponsor's clinical monitoring plan **Checklist Review Interval:** Surveillance imaging (CT/MRI) and ctDNA blood draws every 6 to 12 weeks to monitor for recurrence

15. Sign-off & Approval

Role	Name	Signature	Date					
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					